

Read
Think
Discuss
Apply



Reading Plan Overview

阅读计划总览

Only focus on these required chapters for this stage. / 现阶段只需要看这些必读章节。



Purpose 目的

- ☐ Build conceptual understanding of AI.
建立对人工智能的概念理解。
- ☐ Prepare for NOAI foundation learning.
为 NOAI 基础学习做准备。
- ☐ Read selectively, not the whole book.
有选择地阅读，而不是整本都读。



A
More Thoughtful
You



Required Chapters 必读章节

- ☐ **Chapter 1** The Roots of Artificial Intelligence
人工智能的起源
- ☐ **Chapter 2** Neural Networks and the Ascent of Machine Learning
神经网络与机器学习的崛起
- ☐ **Chapter 5** ConvNets and ImageNet
卷积神经网络与 ImageNet
- ☐ **Chapter 6** A Closer Look at Machines That Learn
更深入地了解机器如何学习
- ☐ **Chapter 7** On Trustworthy and Ethical AI
值得信赖与负责任的人工智能
- ☐ **Chapter 8** Rewards for Robots
为机器人设计奖励



You Do NOT Need 你暂时不需要看

- ☐ Do not try to read the whole book.
不要试图读完整本书。
- ☐ Do not spend time on other chapters yet.
暂时不要花时间读其他章节。
- ☐ Focus on key pages and key ideas.
只聚焦重点页码和核心概念。



Smaller
Reading,
Bigger Learning



Self-Check 自检

- ☐ Do I know which chapters to read?
我知道要读哪些章节吗?
- ☐ Do I know why I am reading them?
我知道为什么读这些内容吗?
- ☐ Will I take notes in my own words?
我会用自己的话做笔记吗?



Check
Reflect
Improve

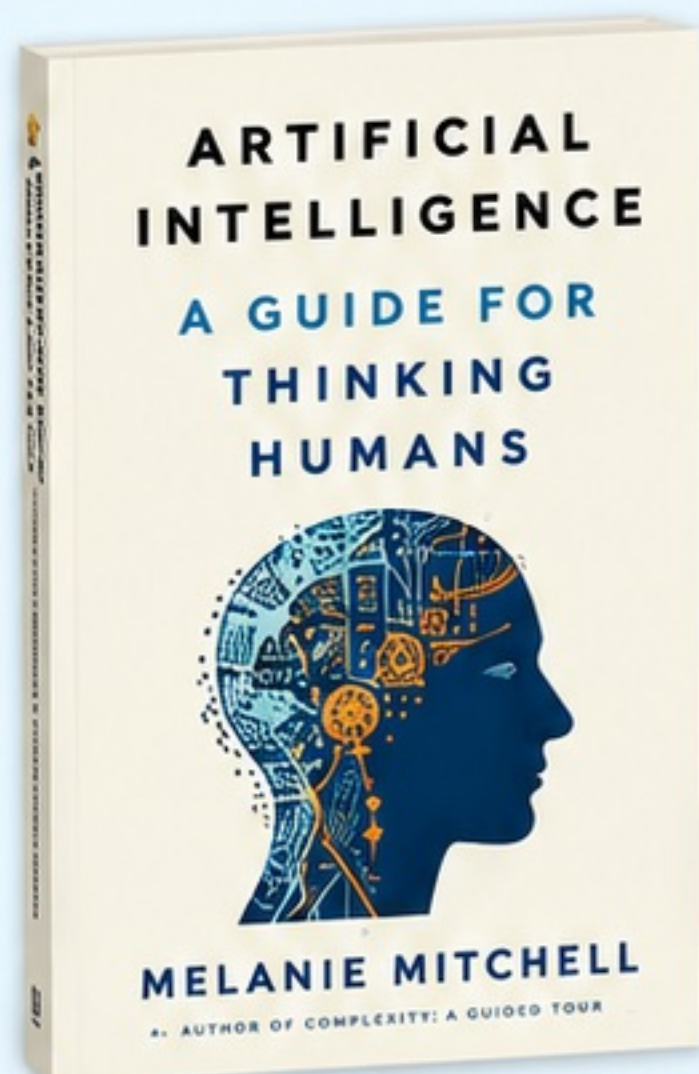


Read only what you need, think clearly, and explain it in your own words.
只读需要的内容，想清楚，并用自己的话讲出来。

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Artificial Intelligence: A Guide for Thinking Humans by Melanie Mitchell

《人工智能：写给思考的人类》 阅读清单



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Build a deeper understanding of AI.
Prepare for NOAI.
Be a more thoughtful human.
建立对人工智能的深层理解。
为 NOAI 做好准备。
成为更有思考力的人。

Knowledge • Curiosity • Responsibility
知识 • 好奇 • 责任

Chapter 1 – The Roots of Artificial Intelligence 第一章：人工智能的起源



1 Key Reading Pages 重点阅读页码

- ☐ pp.17–18 Two Months and Ten Men at Dartmouth
达特茅斯会议
- ☐ pp.19–20 Definitions, and Getting On with It
AI 定义
- ☐ p.20 An Anarchy of Methods
方法的多样性
- ☐ pp.21–23 Symbolic AI
符号主义
- ☐ p.27 Behaviorism paragraph
行为主义相关段落



2 Must-Know Ideas 你必须知道

- ☐ AI does not have one single perfect definition.
AI 没有唯一完美定义。
- ☐ AI has two goals: scientific understanding and practical problem-solving.
AI 有两个目标：科学理解与实际应用。
- ☐ AI is a field, not one single algorithm.
AI 是一个领域，不是一种单一算法。
- ☐ Symbolic AI represents knowledge with symbols and rules.
符号主义用符号和规则表示知识。
- ☐ Behaviorism emphasizes learning through interaction and feedback.
行为主义强调通过交互与反馈学习。



3 Self-Check 自检

- ☐ Can I define AI in one or two sentences?
我能用一两句话定义 AI 吗？
- ☐ Can I explain the two goals of AI?
我能解释 AI 的两个目标吗？
- ☐ Can I explain why there are many AI methods?
我能解释为什么 AI 有多种方法吗？
- ☐ Can I define Symbolism?
我能解释什么是符号主义吗？



4 One-Sentence Memory 记忆一句话



AI is the field of building systems that
perform intelligent tasks, using different
methods for different problems.

AI 是构建能够完成智能任务的系统的领域，
并根据不同问题采用不同方法。



“AI is not just a technology. It's a conversation about what it means to be human.”

AI 不仅仅是一项技术，它也是一场关于何为人類的對話。

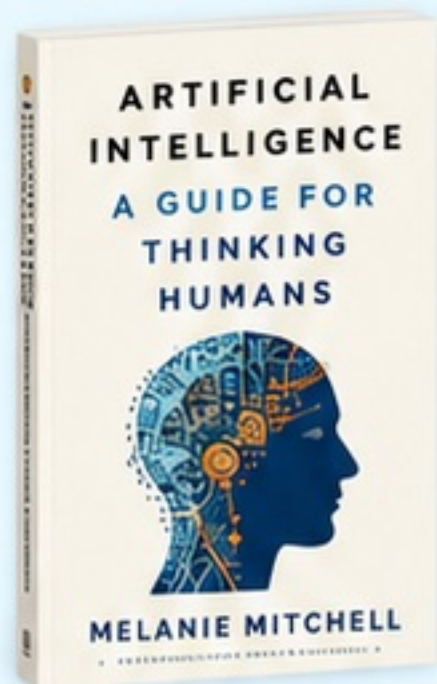
— Melanie Mitchell

— 梅兰妮·米切尔

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Reading Checklist 03

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Read
Think
Discuss
Apply

Chapter 2 — Neural Networks and the Ascent of Machine Learning 第二章：神经网络与机器学习的崛起

Artificial Intelligence: A Guide for Thinking Humans by Melanie Mitchell
《人工智能：写给思考的人类》阅读清单

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1 Key Reading Pages 重点阅读页码



- ☐ pp.35–37 The Return of Neural Networks
神经网络的回归
- ☐ pp.38–40 Learning and Connectionism
学习与连接主义
- ☐ pp.40–41 Connectionism
连接主义
- ☐ p.41 Bad at Logic, Good at Frisbee
不擅长逻辑，但擅长投掷飞盘

2 Must-Know Ideas 你必须知道



- ☐ Connectionism stores knowledge in weighted connections.
连接主义把知识存储在加权连接中。
- ☐ A neural network learns patterns from data.
神经网络从数据中学习规律。
- ☐ Perceptrons are an early model in this tradition.
感知机是这一传统中的早期模型。
- ☐ Connectionism is strong at pattern recognition.
连接主义擅长模式识别。
- ☐ Compared with Symbolism, it learns from data instead of explicit rules.
与符号主义相比，它依靠数据学习，而不是显式规则。

3 Quick Comparison 快速比较



Aspect 比较维度	Symbolism 符号主义	Connectionism 连接主义
How knowledge is stored 知识如何存储	symbols + rules 符号 + 规则	learned weights 学到的权重
Strength 优势	logic and reasoning 逻辑推理	pattern recognition 模式识别

4 Self-Check 自检



- ☐ Can I define Connectionism?
我能解释什么是连接主义吗？
- ☐ Can I explain what a perceptron is at a basic level?
我能基本解释感知机吗？
- ☐ Can I compare Symbolism and Connectionism?
我能比较符号主义和连接主义吗？
- ☐ Can I explain why neural networks became important?
我能解释为什么神经网络变得重要吗？



Remember This
记住这句话

Connectionism learns patterns from data through weighted networks.
连接主义通过加权网络从数据中学习规律。

“AI is not just a technology. It's a conversation about what it means to be human.”

AI 不仅仅是一项技术，它也是一场关于何为人类的对话。

— Melanie Mitchell

—— 梅兰妮·米切尔

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Chapter 5 — ConvNets and ImageNet

第五章：卷积神经网络与 ImageNet

Artificial Intelligence:
A Guide for Thinking Humans

《人工智能：写给思考的人类》阅读清单

Read • Think • Discuss • Apply
阅读 • 思考 • 讨论 • 应用

Learn
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1 Key Reading Pages

重点阅读页码

- ☐ pp.91–95 Early Ideas in Computer Vision
计算机视觉的早期思想
- ☐ pp.96–101 Convolutional Neural Networks
卷积神经网络
- ☐ pp.102–110 ImageNet and Deep Learning
ImageNet 与深度学习
- ☐ pp.110–118 What ConvNets Learn
卷积网络学到了什么



2 Must-Know Ideas

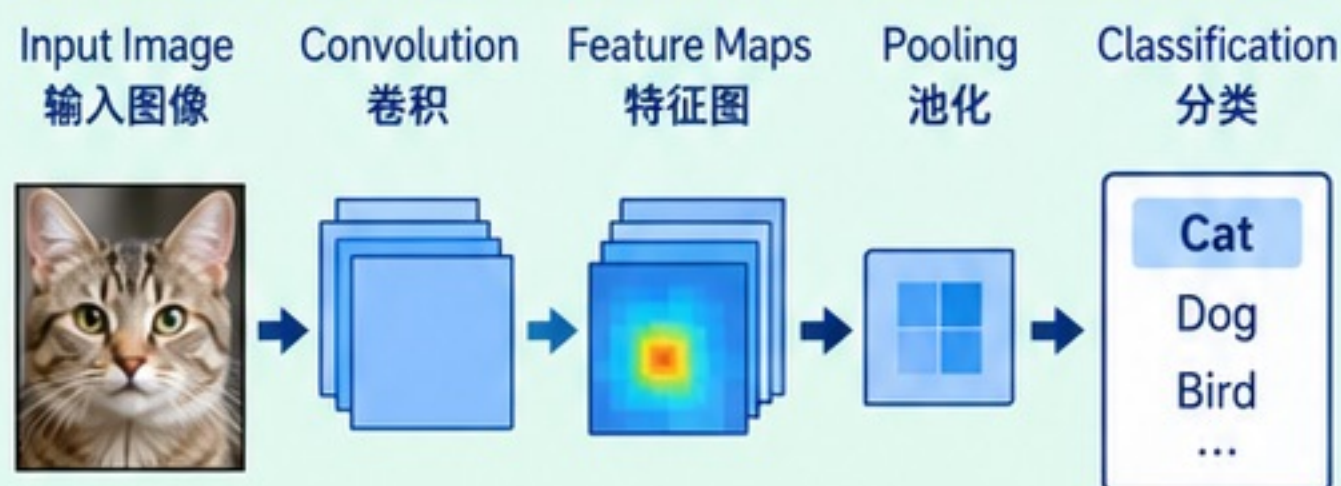
你必须知道

- ☐ CNN stands for Convolutional Neural Network.
CNN 指卷积神经网络。
- ☐ Images are processed through convolutional layers.
图像通过卷积层处理。
- ☐ Pooling helps reduce and summarize features.
池化帮助压缩并概括特征。
- ☐ CNNs learn hierarchical visual features.
CNN 学习分层视觉特征。
- ☐ ImageNet helped drive the deep learning breakthrough.
ImageNet 推动了深度学习突破。



3 Simple Flow

简单流程



4 Self-Check

自检

- ☐ Can I say what CNN means?
我能说出 CNN 的含义吗？
- ☐ Can I explain what pooling does?
我能解释池化的作用吗？
- ☐ Can I explain why ImageNet mattered?
我能解释 ImageNet 为什么重要吗？
- ☐ Can I describe CNN processing in simple steps?
我能用简单步骤描述 CNN 处理过程吗？



Key Memory

重点记忆

CNNs learn visual patterns layer by layer.
CNN 逐层学习视觉模式。

"AI is not just a technology. It's a conversation about what it means to be human."

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Reading Checklist 05

— Learn Together • Think Further • Create a Better Future —

Chapter 6 — A Closer Look at Machines That Learn

第六章：更深入地了解机器如何学习

Artificial Intelligence: A Guide for Thinking Humans
《人工智能：写给思考的人类》



1 Key Reading Pages 重点阅读页码

- ☐ **pp.129–136** How Learning Works
机器学习的工作方式
- ☐ **pp.137–145** Generalization and Overfitting
泛化与过拟合
- ☐ **pp.146–152** Evaluating Models
模型评估
- ☐ **pp.153–160** The Limits of Learning
学习的局限



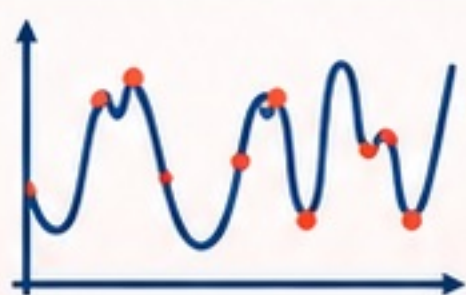
2 Must-Know Ideas 你必须知道

- Training, validation, and test sets play different roles.
训练集、验证集、测试集作用不同。
- Overfitting means memorizing too much.
过拟合意味着记住太多细节。
- Underfitting means the model is too simple.
欠拟合意味着模型过于简单。
- Generalization means performing well on unseen data.
泛化是指在新数据上表现良好。
- A model with high training accuracy can still be bad.
训练集表现高不代表模型就真的好。



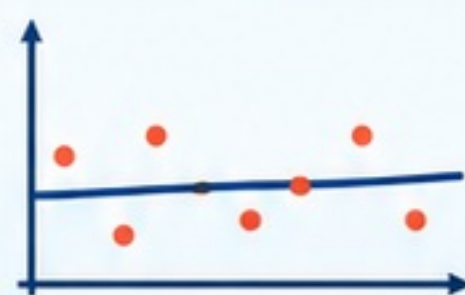
3 Quick Contrast 快速对比

Overfitting 过拟合



Great on training,
weak on new data.
训练好，新数据差。

Underfitting 欠拟合



Weak on both.
两边都差。



4 Self-Check 自检

- ☐ Can I explain train / validation / test?
我能解释训练集、验证集、测试集吗？
- ☐ Can I distinguish overfitting and underfitting?
我能区分过拟合和欠拟合吗？
- ☐ Can I explain generalization?
我能解释泛化吗？
- ☐ Can I explain why evaluation matters?
我能解释为什么模型评估重要吗？



Remember
记住

A good model learns patterns, not just the training set.
好模型学的是规律，而不只是训练集。

“AI is not just a technology. It's a conversation about what it means to be human.”

AI 不仅仅是一项技术，它也是一场关于何为人类的对话。 — Melanie Mitchell

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Chapter 7 — On Trustworthy and Ethical AI 第七章：值得信赖与负责任的人工智能

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For a
More Human
AI Future



1 Key Reading Pages 重点阅读页码

- ☐ pp.161–168 Bias in AI
AI 中的偏见
- ☐ pp.169–178 Privacy and Surveillance
隐私与监控
- ☐ pp.179–189 Responsibility
责任
- ☐ pp.190–200 Toward Trustworthy AI
迈向可信赖的 AI



2 Must-Know Ideas 你必须知道

- AI systems can reflect bias in data.
AI 系统会反映数据中的偏见。
- Data privacy matters.
数据隐私很重要。
- Surveillance raises ethical concerns.
监控会带来伦理问题。
- Humans are still responsible for AI use.
人类仍需对 AI 的使用负责。
- Trustworthy AI requires care, fairness, and accountability.
可信赖的 AI 需要谨慎、公平和问责。



3 What To Ask 你要会问

- ☐ Who collected the data?
谁收集了数据？
- ☐ Could the data be biased?
数据可能有偏见吗？
- ☐ Who might be harmed?
谁可能受到伤害？
- ☐ How can risk be reduced?
如何降低风险？



4 Self-Check 自检

- ☐ Can I explain algorithmic bias?
我能解释算法偏见吗？
- ☐ Can I explain why privacy matters?
我能解释为什么隐私重要吗？
- ☐ Can I explain responsibility in AI?
我能解释 AI 责任吗？
- ☐ Can I suggest one way to reduce harm?
我能提出一种减少伤害的方法吗？



Remember 记住

Powerful AI must also be fair, safe, and responsible.
强大的 AI 也必须公平、安全、负责任。

“AI is not just a technology. It’s a conversation about what it means to be human.”

AI 不仅仅是一项技术，它也是一场关于何为人類的對話。”

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Chapter 8 — Rewards for Robots 第八章：为机器人设计奖励

Read • Think • Discuss • Apply
阅读 • 思考 • 讨论 • 应用

Smarter
People
Brighter
Tomorrows



LEARN TOGETHER
CREATE A BETTER FUTURE

1 Key Reading Pages 第点阅读页码



- ☐ pp.201–210 What Is Reinforcement Learning?
什么是强化学习？
- ☐ pp.211–220 Learning Through Reward and Punishment
通过奖励与惩罚学习
- ☐ pp.221–234 Applications and Challenges
应用与挑战

2 Must-Know Ideas 你必须知道



- Reinforcement learning involves an agent and an environment.
强化学习涉及智能体和环境。
- The agent takes actions and receives rewards.
智能体采取行动并获得奖励。
- Learning happens over states, actions, and rewards.
学习围绕状态、动作和奖励展开。
- Exploration and exploitation are both important.
探索与利用都很重要。
- Reinforcement learning is useful in games and robotics.
强化学习在游戏和机器人中很有用。

3 Simple Loop 简单循环



4 Self-Check 自检



- ☐ Can I define reinforcement learning?
我能定义强化学习吗？
- ☐ Can I name agent, action, environment, and reward?
我能说出智能体、动作、环境和奖励吗？
- ☐ Can I explain exploration vs. exploitation?
我能解释探索与利用吗？
- ☐ Can I give one real-world application?
我能举一个真实应用吗？



Remember This
记住这一点

Reinforcement learning improves behavior
through interaction and reward.
强化学习通过交互和奖励改进行为。

Small Rewards
Big Possibilities



Chapter 3 — AI Spring

第三章：AI 春天

Read · Think · Discuss · Apply | 阅读 · 思考 · 讨论 · 应用



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Purpose 目的

- ☐ Understand the current wave of AI development and why it is called an “AI Spring.”
了解当前的 AI 发展浪潮，以及为什么被称为 “AI 春天”。
- ☐ See how new technologies, data, and computing power came together to drive rapid progress.
了解新技术、数据和算力如何结合，推动了快速的进步。
- ☐ Recognize the opportunities and challenges that come with this AI Spring.
认识到 AI 春天带来的机遇和挑战。



Required Chapters 必读章节

- ☐ **Chapter 3** AI Spring
AI 春天
- ☐ **Chapter 4** Transformers and a New Era
Transformer 与新时代
- ☐ **Chapter 5** Large Models Change the World
大模型改变世界
- ☐ **Chapter 6** AI in Science and Medicine
AI 在科学与医疗中的应用



You Do NOT Need 你暂时不需要看

- ☐ Do not try to read the whole book at once.
不要试图一次性读完整本书。
- ☐ Do not spend time on later chapters yet.
暂时不要把时间花在后面的章节。
- ☐ Focus on key pages and key ideas for this stage.
只聚焦这个阶段的关键页面和核心概念。



Smaller
Reading,
Bigger Learning



Self-Check 自检

- ☐ Do I understand why this period is called an “AI Spring”?
我是否理解为什么这个时期被称为 “AI 春天”？
- ☐ Can I explain the key factors (technology, data, and computing power) behind recent AI progress?
我能解释近期 AI 进步背后的关键因素（技术、数据和算力）吗？
- ☐ Can I identify the main opportunities and challenges of this AI Spring?
我能识别这个 AI 春天的主要机遇和挑战吗？



Check
Reflect
Improve



Read only what you need, think clearly, and explain it in your own words.
只读需要的内容，想清楚，并用自己的话讲出来。

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Chapter 9 — Game On

第九章：游戏开始

Read · Think · Discuss · Apply | 阅读 · 思考 · 讨论 · 应用



1 Key Reading Pages 重点阅读页码

- ☐ pp.221–222 Games as a Testbed
游戏作为试验场
- ☐ pp.223–225 Learning and Skills
学习与技能
- ☐ pp.226–228 Real-World Impact
现实世界的影响
- ☐ pp.229–230 What Comes Next
接下来会发生什么
- ☐ pp.231–233 Chapter Summary
本章总结



2 Must-Know Ideas 你必须知道

- ☐ Games provide a safe space to test and improve AI.
游戏为测试和改进 AI 提供了一个安全的环境。
- ☐ Playing games helps AI learn through trial and error.
通过玩游戏，AI 可以在试错中学习。
- ☐ AI can develop complex skills in simulated environments.
AI 可以在模拟环境中发展复杂的技能。
- ☐ Success in games can lead to real-world applications.
在游戏中取得的成功可以带来现实世界的应用。
- ☐ Games show us both the potential and limits of AI.
游戏向我们展示了 AI 的潜力和局限性。



3 What To Ask 你要会问

- ☐ What makes a game a good test for AI?
什么样的游戏是 AI 的良好测试场？
- ☐ What can AI learn from playing games?
AI 可以从玩游戏中学到什么？
- ☐ How do game skills transfer to real-world tasks?
游戏中学到的技能如何应用到现实任务中？
- ☐ What are the benefits and risks of using games for AI development?
将游戏用于 AI 开发有哪些好处和风险？



4 Self-Check 自检

- ☐ Can I explain why games are useful for AI research?
我能解释为什么游戏对 AI 研究有用吗？
- ☐ Can I give examples of skills AI learns in games?
我能举出 AI 在游戏中学到的技能的例子吗？
- ☐ Can I explain how game research might help in real life?
我能解释游戏研究如何在现实生活中提供帮助吗？
- ☐ Can I share my thoughts on the future of AI in games?
我能分享我对游戏中 AI 未来发展的看法吗？



Remember 记住

Games today can build a smarter, more helpful tomorrow.
今天的游戏可以创建一个更智能、更有帮助的明天。



Chapter 11 — Words, and the Company They Keep

第十一章：词语及其伙伴

Read · Think · Discuss · Apply | 阅读 · 思考 · 讨论 · 应用



1 Key Reading Pages

重点阅读页码

- ☐ pp.215–217 Why words have meaning
为什么词语有意义
- ☐ pp.218–221 Distributional semantics
分布式语义
- ☐ pp.222–224 Word embeddings
词嵌入
- ☐ pp.225–227 Analogies and semantic relationships
类比与语义关系



2 Must-Know Ideas

你必须知道



- ☐ Words get their meaning from the contexts they appear in.
词语通过它们出现的上下文获得意义。
- ☐ Distributional semantics captures meaning from word co-occurrence.
分布式语义通过词语的共现来捕捉意义。
- ☐ Word embeddings represent words as vectors in a high-dimensional space.
词嵌入将词语表示为高维空间中的向量。
- ☐ Words with similar contexts tend to have similar meanings.
具有相似上下文的词语往往具有相似的意义。



3

Quick Comparison

快速比较



Aspect 比较维度	Traditional View 传统观点	Distributional View 分布式观点
Meaning source 意义来源	human-defined rules 人工定义的规则	context and co-occurrence 上下文与共现
Representation 表示方式	symbols 符号	dense vectors 稠密向量
Similarity 相似性	manual grouping 人工分类	learned from data 通过数据学习



4

Self-Check

自检



- ☐ Can I explain why words have meaning?
我能解释为什么词语有意义吗？
- ☐ Can I describe what distributional semantics is?
我能描述什么是分布式语义吗？
- ☐ Can I explain how word embeddings work?
我能解释词嵌入是如何工作的吗？
- ☐ Can I give an example of an analogy between words?
我能举一个词语类比的例子吗？



Remember This
记住这句话

Meaning emerges from patterns of use.
意义源于使用的模式。



Chapter 12 — Translation as Encoding and Decoding

第十二章：编码与解码式翻译

Read · Think · Discuss · Apply | 阅读 · 思考 · 讨论 · 应用

**1**

Key Reading Pages

重点阅读页码

- ☐ **pp.291–295** Introduction to Neural Machine Translation
神经机器翻译导论
- ☐ **pp.296–300** The Encoder-Decoder Architecture
编码器-解码器架构
- ☐ **pp.301–305** Training and Inference in Machine Translation
机器翻译的训练与推理
- ☐ **pp.306–310** Evaluation and Applications of Machine Translation
机器翻译的评估与应用

**2**

Must-Know Ideas

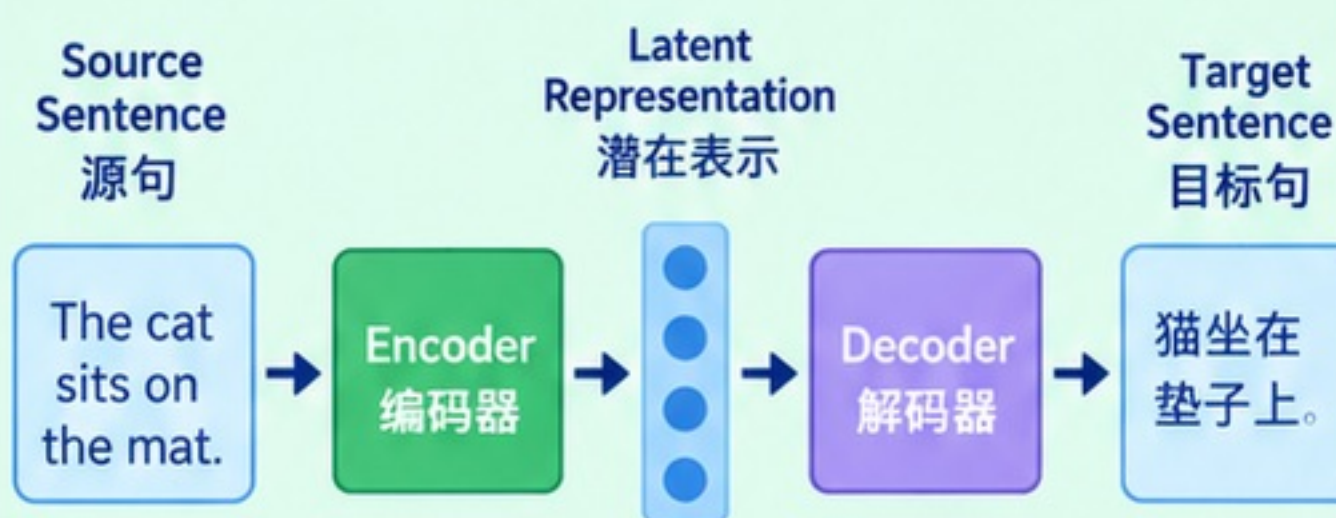
你必须知道

- ☐ Translation can be viewed as an encoding and decoding process.
翻译可以被看作是一个编码与解码的过程。
- ☐ Neural machine translation uses an encoder-decoder architecture.
神经机器翻译采用编码器-解码器架构。
- ☐ Attention helps the model focus on relevant source information.
注意力机制帮助模型关注相关的源语言信息。
- ☐ Evaluation metrics such as BLEU are used to measure translation quality.
常用 BLEU 等评估指标来衡量翻译质量。

**3**

Simple Flow

简单流程

**4**

Self-Check

自检

- ☐ Can I explain what encoding means in translation?
我能解释翻译中编码是什么意思吗？
- ☐ Can I explain what decoding means in translation?
我能解释翻译中解码是什么意思吗？
- ☐ Can I describe the role of attention in machine translation?
我能描述注意力机制在机器翻译中的作用吗？
- ☐ Can I compare different evaluation metrics for translation?
我能比较不同的翻译评估指标吗？



Key Memory

重点记忆

Translation is an encoding and decoding process.
翻译是一个编码与解码的过程。